

SIFT MEDIA

Full system report

Every source, every gate, every language, every end message.

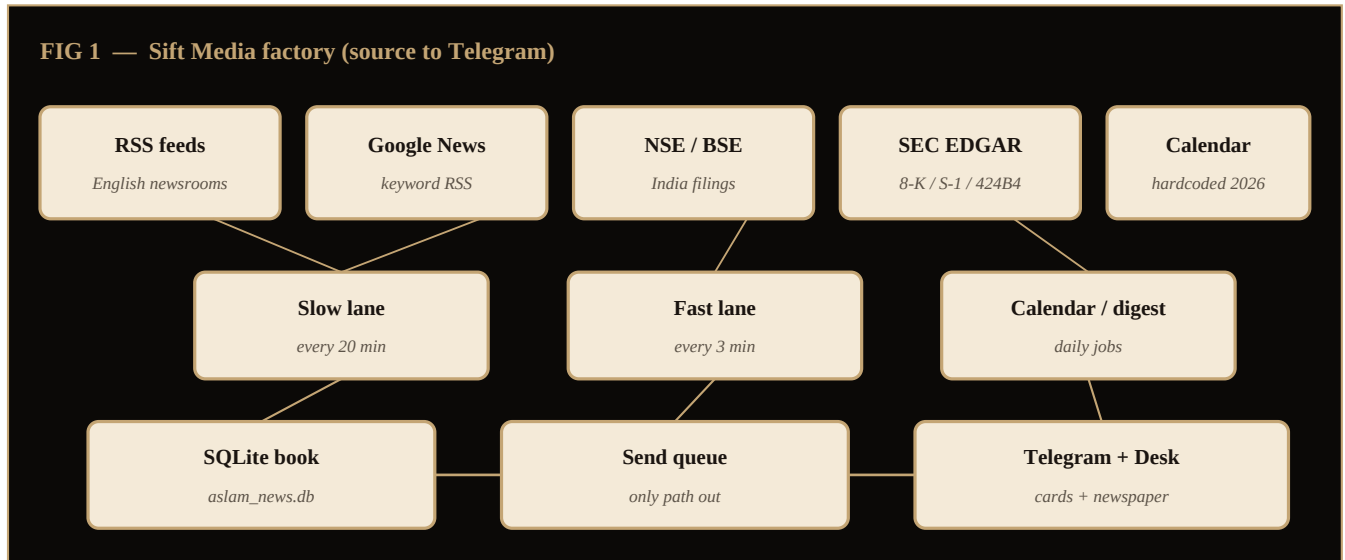
Drawn from live code on 13 September 2026. Printed Sunday, 13 September 2026, 04:18 PM IST.

Public brand is **Sift Media**. Folders on disk are still `/root/aslam-news-media` (bot) and `/root/aslam-media-desk` (website). Telegram channel names and the BotFather username were not renamed.

1. What this project is

Sift Media is a news factory, not a newsroom with writers. A Python service pulls public feeds, throws most items away, asks a language model to keep only serious stories, then posts a short English card to Telegram. A website called Live Desk is a newspaper view of the same book. It never fetches news itself.

There is one running bot (single-bot mode: BOT2_TOKEN is empty). systemd unit `aslam-news-media.service` keeps it alive. The desk is a separate process on port 8780, not systemd.



2. Runtime and stack

Where it lives

Piece	Fact
VPS	CloudOnFire KVM vpsaevm9ub75 · public IP 160.250.224.8 · Linux
Bot code	/root/aslam-news-media · Python 3 venv
Website	/root/aslam-media-desk · port 8780
Database	/root/aslam-news-media/data/aslam_news.db (SQLite)
Logs	/root/aslam-news-media/data/bot.log
Service	aslam-news-media.service (enabled, restart=always)
Desk HTTPS	Cloudflare quick tunnel (hostname dies on reboot)
Desk HTTP	http://160.250.224.8:8780

Languages of the system (this is the language section)

People language and machine language are different. Both are English-first.

Layer	Language / format
Human output (Telegram, desk, PDF)	English only. No Hindi/Tamil generation in this bot.
RSS / Google News	English publishers. Google queries use hl=en-US, gl=US, ceid=US:en.
HTTP headers to NSE/BSE	Accept-Language: en-US,en;q=0.9
Calendar times	US prints shown as ET + IST. IST in AM/PM for chat answers.
Live Desk clock	IST, 12-hour style on the API clock string.

Layer	Language / format
Code	Python 3. Libraries: aiogram 3, APScheduler, feedparser, rapidfuzz, openai SDK, httpx, requests, nsepython (present), reportlab, pymupdf.
Telegram markup	HTML parse mode. Title is the hyperlink. Link preview off.
LLM JSON	English keys: keep, decision, summary_ai, why_matters, watch_next, significance.
Not this project	The IG Tamil/Hinglish translator bot is a different repo (/root/ig-tamil-bot).

Python packages (requirements.txt)

```
aiogram>=3.13      feedparser>=6.0      APScheduler>=3.10
rapidfuzz>=3.9     openai>=1.40         httpx>=0.27
requests>=2.31     python-dotenv         nsepython>=0.1.9
reportlab>=4.0     pymupdf>=1.24
```

3. The clock: what runs, how often

Job	When (UTC)	IST	What it does
slow_lane	every 20 min	—	RSS + Google News → news cards
fast_lane	every 3 min	—	NSE + BSE + SEC → market alerts
health	every 60 min	—	If a fetcher goes stale, ping admin
calendar	08:00 UTC	13:30 IST	T-3 explainers + TODAY pings
digest_morning	02:30 UTC	08:00 IST	PDF of last ~14h kept stories
digest_evening	13:30 UTC	19:00 IST	PDF of last ~11h kept stories
boot kick	on start	—	One slow + one fast cycle immediately

APScheduler is AsyncIOScheduler. Jobs are coroutine functions (not lambdas). A crash in one job cannot kill the scheduler. Both lanes start once at boot so you do not wait a full interval after a restart.

4. Every source we use

No paid data vendor. No GDELT, Moneycontrol, or Finnhub (those flags stay off until you say GO). Full article text is never stored. Only title, a short snippet (cap 280 characters), source, category, and link.

4.1 Slow lane — RSS newsrooms (English)

Internal name	Publisher	Category	Feed
coindesk	CoinDesk	crypto_news	coindesk.com RSS
cointelegraph	Cointelegraph	crypto_news	cointelegraph.com/rss
decrypt	Decrypt	crypto_news	decrypt.co/feed
cnbc_markets	CNBC	us_stock_market	CNBC markets RSS
cnbc_finance	CNBC	us_stock_market	CNBC finance RSS
marketwatch_top	MarketWatch	us_stock_market	Dow Jones MW top stories
yahoo_finance	Yahoo Finance	us_stock_market	finance.yahoo.com RSS
npr_politics	NPR	us_politics	NPR politics RSS
thehill	The Hill	us_politics	thehill.com feed
et_markets	Economic Times	india_market_ipo	ET markets RSS
et_stocks	Economic Times	india_market_ipo	ET stocks RSS
et_ipo	Economic Times	india_market_ipo	ET IPO RSS
livemint_markets	Livemint	india_market_ipo	livemint.com/rss/markets
bs_markets	Business Standard	india_market_ipo	BS markets RSS

Internal name	Publisher	Category	Feed
thehindu_business	The Hindu	india_market_ipo	thehindu.com business RSS
ndtv_india	NDTV	india_politics	NDTV India feed
toi_india	Times of India	india_politics	TOI India RSS
bbc_mideast	BBC	geopolitics	BBC Middle East RSS
dw_world	DW	geopolitics	DW world RDF
france24_world	France 24	geopolitics	france24.com/en/rss
thediplomat	The Diplomat	geopolitics	thediplomat.com/feed
reuters_world_gnews	Reuters via GNews	geopolitics	Google News site:reuters.com world
bbc_world	BBC	daily_news	BBC world RSS
aljazeera	Al Jazeera	daily_news	aljazeera.com RSS
reuters_via_gnews	Reuters via GNews	daily_news	Google News site:reuters.com 24h

Business Standard is known to serve broken XML to custom User-Agents and valid XML to a browser UA. The RSS fetcher uses a browser UA and retries the other UA if the feed is empty.

4.2 Slow lane — Google News keyword searches (unofficial, English)

These are news.google.com/rss/search URLs. Unofficial, redirect-encoded links, about 100 items cap. Links are unwrapped when possible.

Name	Query (when:1d unless noted)	Category
gnews_gold	gold price OR bullion OR XAU	gold_news
gnews_crypto	bitcoin OR ethereum OR crypto regulation	crypto_news
gnews_us_ipo	IPO pricing OR IPO debut OR S-1 filing	us_market_ipo
gnews_india_ipo	India IPO OR SME IPO OR NSE listing	india_market_ipo
gnews_fed	Federal Reserve OR FOMC OR US inflation CPI	us_econ_calendar
gnews_rbi	RBI policy OR India CPI OR India GDP	india_econ_calendar
gnews_us_pol	US Congress OR White House policy	us_politics
gnews_in_pol	India government policy OR Parliament session	india_politics
gnews_geo_conflict	ceasefire OR airstrike OR military offensive OR border clash	geopolitics
gnews_geo_diplomacy	sanctions OR peace talks OR NATO OR UN Security Council	geopolitics
gnews_geo_asia	Taiwan OR South China Sea OR North Korea tensions	geopolitics
gnews_geo_india	India Pakistan OR India China border OR India foreign policy	geopolitics

4.3 Fast lane — NSE India (keyless, cookie session)

Host <https://www.nseindia.com/api/...> Browser User-Agent, cookie bootstrap every 4 minutes, 1 second sleep between feeds. Shareholding-pattern JSON was removed by NSE (404s) so it is not scheduled. Shareholding still can appear inside announcements.

Fetcher	What	Event type
nse_deals	Bulk + block deals	bulk_deal / block_deal
nse_announcements	Corporate announcements	announcement
nse_ipo	Mainboard + SME IPO	ipo
nse_corp_actions	Dividend / bonus / split etc.	corp_action
nse_results	Quarterly financial results (top 50)	earnings
nse_fii_dii	FII/DII daily rupee flows	fii_dii
nse_circulars	Exchange circulars (top 40)	circular

4.4 Fast lane — BSE India (secondary, keyless)

api.bseindia.com with browser Referer/Origin. Shape is messy (object, double-encoded JSON, or 'No Record Found!'). Empty on weekends is treated as success so health does not false-alarm. Used as a cross-check; NSE is primary. Dedup later via event_key + fuzzy title.

4.5 Fast lane — US SEC EDGAR (official, free)

Form	Category	Event	Endpoint
8-K	us_stock_market	earnings / material event	sec.gov browse-edgar atom, 40 latest
S-1	us_market_ipo	ipo	same, type=S-1
424B4	us_market_ipo	ipo (priced)	same, type=424B4

Needs a descriptive User-Agent (config SEC_EDGAR_UA). Throttled under the public 10 req/s limit. Empty boilerplate S-1 titles and empty 8-Ks are rule-rejected later.

4.6 Economic calendar (not a live scrape)

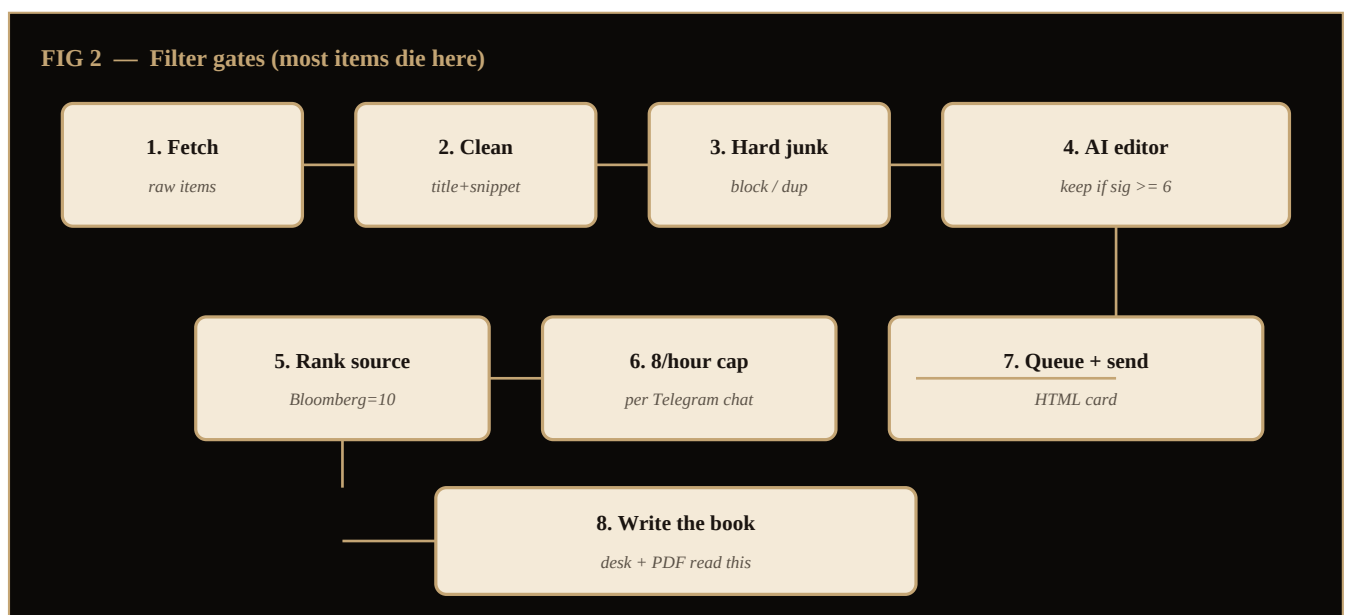
Dates are a hardcoded 2026 catalog in fetch/econ_calendar.py (BEA, BLS, FOMC, MoSPI, RBI). It is not a live BLS pull. Horizon is rolling 30 days. Same-day prints are included. GDP and PCE at the same timestamp are two rows. Empty event_type rows are skipped (old dummy rows used to fire false TODAY pings).

US series in the catalog include NFP, CPI, PPI, PCE, GDP estimates, retail sales, ISM PMI, JOLTS, FOMC, Labor Day. India: CPI, IIP, GDP, RBI MPC. T-3 = full card (why it matters + markets / gold / BTC mechanism, no invented numbers, no buy/sell). TODAY = short ping if importance ≥ 4 . Chat answers after a print should show Actual vs Forecast vs Previous when known; if Investing/TE/FF show a blank Actual, the bot must say blank, not invent a tweet.

4.7 Explicitly NOT used

- GDELT, Moneycontrol, Chittorgarh, Finnhub (ENABLE_* flags default 0).
- Paid FMP / FRED.
- Full-page scrapes or article body storage.
- X/Twitter guest-token tricks (out of scope for this bot).
- Varaha desk on :8766 (equity picks, different product).

5. Path of one news item (slow lane)



A typical slow cycle pulls 1,000+ raw items and may queue under 10 cards.

- `rss.fetch_all_rss()` hits every RSS + Google News query (fail-soft per feed).
- `pipeline.normalize`: strip HTML, canonical URL, 280-char snippet, keyword category guess. Geopolitics strong-words (airstrike, sanctions, NATO...) beat 'Trump' so a strike on another country is not US domestic politics.
- `pipeline.prefilter`: URL hash already in `seen_urls` → drop. `rapidfuzz` title near-duplicate in 72h window → drop. Blocked domains/names → drop and mark seen. Clickbait regex → drop.
- Newest first, then cap `MAX_ITEMS_PER_CYCLE` (40) before the AI.
- `smart_filter.filter_batch` in groups of 10. JSON keep/drop + `summary_ai` / `why_matters` / `watch_next` / significance.
- Keep only if the model says keep AND significance ≥ 6 (config `POST_MIN_SIGNIFICANCE`). Note: the prompt text still says 5 in one place; the code floor is 6.
- `db.mark_seen` BEFORE send. If this chat already posted 8 times in the last hour, skip.
- `send_queue.enqueue` HTML card. `db.mark_posted` after enqueue. Rejects also mark_seen so they are never re-judged.

6. Path of one market item (fast lane)

- `collect()` runs SEC, then NSE, then BSE. One crash cannot stop the others.
- Collapse bulk/block: one card per symbol per cycle, largest notional. Skip under ■10 crore.
- Rule-reject without AI: newspaper publication, director change, empty analyst meet, Record Date, empty S-1/8-K boilerplate. Keep dividends, bonus, buyback, order wins, earnings-with-numbers, priced NSE IPOs, material 8-K.
- Each remaining item is its own LLM call (3-way: keep / reject / review). Workers capped (`FAST_FILTER_WORKERS`, env 2).
- keep → public category channel after chat cap. review → admin chat, separate cap 4/hour. reject → silent, still recorded.
- `posted_at` on market rows is set only after a real send (otherwise the cap lies).

If the LLM is down

Slow lane fail-soft: keep-wire-only (reliability ≥ 8), not keep-all. Fast lane fail-soft: review, never auto-post an unjudged market item. systemd 'active' is not proof the editor is alive. Check the tail of `bot.log`.

7. Source ranks (hardcoded, never the LLM)

Missing brand = 5. That 5 is a hole, not a grade. Google News 'Bloomberg.com' plus a `news.google.com` URL must still score 10 (TLD strip + ignore Google as host).

Score	Who
10	Bloomberg, SEC EDGAR
9	Reuters, AP, AFP, BBC, WSJ, FT, NSE, BSE
8	TOI, NDTV / NDTV Profit, ET, Livemint, BS, Hindu, HT, Indian Express, CNBC / TV18, Forbes / Forbes India, Barron's, NYT, WaPo, Telegraph, Guardian, Al Jazeera, DW, France 24, NPR, Diplomat
7	MarketWatch, The Hill, CoinDesk, Cointelegraph, Decrypt, Investing.com / Investing.com India
6	Yahoo Finance, TheStreet
5	Anything not in the table (even a famous masthead we forgot)
0 block	Motley Fool, Insider Monkey, 24/7 Wall St, Barchart, Benzinga, Seeking Alpha

8. The end message (what Telegram actually shows)

The publisher's sentences are not copied. The body is the model's own words (or a short snippet fallback). The title is the hyperlink. Raw URLs are hidden.

Full card (most categories)

● Significance: 7/10 · ■ Gold

Gold pushed to a fresh record as traders priced in a rate cut ← title = link

■ Reuters · ■■ Reliability 9/10

■ 13 Sep 2026, 04:10 PM IST

■ What happened

...summary_ai, 1-3 sentences, own words...

■ Why it matters

...mechanism, not vibes. No buy/sell...

■ Watch next

...one concrete next print or event...

■ Read more · reuters.com

Daily News uses a compact card (that channel used to drown in volume). Market cards add an event_type line (bulk_deal, ipo, 8-K). If Telegram rejects the HTML, the queue strips tags and resends plain text once.

Telegram category labels

DB category	Label on the card	Typical channel
crypto_news	crypto	CHANNEL_CRYPTONNEWS
geopolitics	geopolitics	CHANNEL_GEOPOLITICS
gold_news	Gold	CHANNEL_GOLDNEWS
india_politics	India Politics	CHANNEL_INDIA_POLITICS
us_politics	US Politics	CHANNEL_US_POLITICS
india_market_ipo	India Market / IPO	CHANNEL_INDIA_MARKET_IPO
us_stock_market	US Stock Market	CHANNEL_US_STOCK_MARKET
us_market_ipo	US Market / IPO	falls back to default if unset
india_econ_calendar	India Economic Calendar	CHANNEL_INDIA_ECON_CALENDAR
us_econ_calendar	US Economic Calendar	CHANNEL_US_ECON_CALENDAR
daily_news	Daily News	default / SLOW_LANE_CHANNEL_ID
admin_health	Admin / Health	ADMIN_DM_ID

Unset categories fall back to SLOW_LANE_CHANNEL_ID. Review + health go to ADMIN_DM_ID. Channel numeric IDs live in .env (not printed here).

Send queue limits (the only path to Telegram)

Limit	Value	Why
Global	~25 / sec	Telegram hard cap ~30/s
Per chat	~1 / sec	do not burst a group
Per chat / min	19	groups cap at 20/min
Editorial / hour	8 public posts per chat	news + markets share this
Admin review / hour	4	stop review dumps
RetryAfter	honoured	Telegram flood control
Priority	0 alerts, 5 news	calendar/health jump the queue

9. The book (database)

Table	Role
seen_urls	Every slow-lane URL. Hash unique. Stores title, AI fields, significance, posted_at, lane.
market_events	Fast-lane events. event_key unique. decision keep/reject/review. posted_at only after send.
calendar_events	Seeded 30-day catalog. UNIQUE(date, event_name, country). alerted_at / dayof_alerted_at.
fetcher_last_run	Health: last success, last error, items_seen, expected_min, lane.
health_alerts	When we already pinged admin about a dead fetcher.

URL identity is SHA-256 of the canonical URL. Mark seen before send so a crash cannot double-post. Desk opens the file as SQLite mode=ro.

10. The website (Live Desk)



The website is not a second factory. It is a window.

Code: backend.py (stdlib ThreadingHTTPServer) + index.html + style.css + app.js. Masthead is now **Sift Media · LIVE DESK · IST** clock. Nav: FRONT, DAILY, WORLD, GOLD, INDIA CALENDAR, INDIA MARKETS, INDIA POLITICS, US CALENDAR, US POLITICS, US MARKETS. Tabs WIRE / DIGEST. The N/10 kicker is source reliability, not LLM significance (often NULL). API /api/state JSON, refresh about every minute. Phone: swipe nav, 44px taps, stack on small screens. CSP is strict (self only). Not systemd — after a VPS reboot, start backend.py again. trycloudflare hostnames die on reboot; IP:8780 is the backup.

11. PDF digest

No extra LLM call. Reads stored summary_ai / why_matters / watch_next. Masthead now SIFT MEDIA. Morning window ~14h, evening ~11h, significance floor 6, skip the file if fewer than 4 items. Sent as a Telegram document via a throwaway Bot then session.close(). Must not call asyncio.run() from the live scheduler loop (that bug sent 0 PDFs). Must not close the shared send-queue session.

12. AI editor (current wiring)

Knob	Live value 13 Sep 2026
Provider actually used	b.ai https://api.b.ai/v1
Model	deepseek-v4-flash-vision-exp (reasoning + vision model used as text JSON filter)
OpenRouter on this bot	NOT wired. No OPENROUTER_API_KEY in aslam-news-media/.env
Wallet	b.ai balance=0 → HTTP 400 insufficient_user_quota
Fail-soft seen in log	slow keep-wire-only; fast keep: 0
Timeout / max tokens	120s / 8000 (reasoning budget)
Pace	LLM_MIN_GAP_SEC 3.5 (b.ai 429s if concurrent)
Original spec	OpenRouter deepseek/deepseek-v4-flash — never landed on this disk after VPS move

OpenRouter free models ARE used by the separate Manus bot (/root/veritas-manus), not by Sift. Switching Sift needs an explicit GO: copy the key, point base URL at openrouter.ai, pick a :free model that can emit JSON, restart the service, prove one keep/reject.

13. Files map

File	Job
main.py	Scheduler + CLI (run / once / health / calendar / digest)
config.py	All knobs from env. Missing cred disables a feature, never crashes.
feeds.py	RSS + Google News list
fetch/rss.py, nse.py, bse.py, sec_edgar.py, econ_calendar.py	Collectors
pipeline.py	Clean, category, ranks, prefilter, HTML cards, channel route
smart_filter.py	LLM JSON keep/drop + explainers
bots/slow_lane.py, fast_lane.py	One cycle each
send_queue.py	Only Telegram exit
db.py	SQLite + migrations
daily_digest.py	PDF
monitor.py	Health + calendar alerts
aslam-media-desk/*	Read-only newspaper

14. Live status at print time

- Bot systemd: active (running since 11 Sep 2026).
- Fetchers: NSE and SEC still pulling (hundreds of NSE items per fast cycle).
- LLM filter: dead (b.ai wallet 0).
- Live Desk: HTTP 200, brand Sift Media, ~9,400 posts in the book.
- Public name locked: Sift Media. Folders/channels still aslam-*
- GitHub PAT is on the VPS for aslam9949; this project is not a git repo and was not pushed.

15. One-sentence memory

English public feeds go in. Rules kill junk. An editor model (currently offline) keeps only significance 6+. At most 8 cards an hour leave through one queue. Telegram is the publisher. SQLite is the book. The website is the newspaper of that book.

End of report. No API keys, chat IDs, or tokens are printed here.